

A Compliance Checking Framework Based on Retrieval Augmented Generation

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Abstract

The text-based compliance checking aims to verify whether a company's business processes comply with laws, regulations, and industry standards using NLP techniques. Existing methods can be divided into two categories: Logic-based methods offer the advantage of precise and reliable reasoning processes but lack flexibility. Semantic embedding methods are more generalizable; however, they may lose structured information and lack logical coherence. To combine the strengths of both approaches, we propose a compliance checking framework based on Retrieval-Augmented Generation (RAG). This framework includes a static layer for storing factual knowledge, a dynamic layer for storing regulatory and business process information, and a computational layer for retrieval and reasoning. We employ an eventic graph to structurally describe regulatory information as we recognize that the knowledge in regulatory documents is centered not on entities but on actions and states. We conducted experiments on Chinese and English compliance checking datasets. The results demonstrate that our framework consistently achieves state-of-the-art results across various scenarios, surpassing other baselines.

1 Introduction

Compliance checking is a critical tool that ensures a company's operations adhere to relevant laws, regulations, and standards, helping prevent violations (Esposito et al., 2023; Robaldo et al., 2024), reduce legal risks, and support sustainable development. Text-based compliance checks focus on leveraging natural language processing techniques to analyze business process documents and regulatory documents, aiming to uncover potential violations (Cejas et al., 2023; Fitkau and Hartmann, 2024; Ren et al., 2024).

The left side of Figure 1 illustrates a schematic of text-based compliance checking. The second sentence in the business process document posits that users, upon registration, consent to share their location information with partners for service provision, implying acceptance of certain data-sharing terms without explicit agreement. However, the third statement in the General Data Protection Regulation (GDPR) unequivocally asserts that user data shall not be shared with third parties unless explicit consent is granted. Thus, the operational protocol of this APP encounters a compliance conflict.

Text-based compliance checking methods can be broadly categorized into two types: logic reasoning methods (Bhuiyan et al., 2024; Fitkau and Hartmann, 2024), as illustrated in the upper right corner of Figure 1, and semantic embedding methods (Beach et al., 2024; Chen et al., 2024), as depicted in the lower right corner of Figure 1. Logic reasoning methods typically utilize tools such as Petri nets, first-order predicate logic, and BPMN to describe business process information and regulatory information. Compliance is then determined through explicit logical reasoning. These methods provide precise inference rules to reduce ambiguity and improve checking accuracy but often lack scalability, making adaptation to changing business environments difficult. Semantic embedding methods represent the semantics of business processes and regulatory requirements using low-dimensional continuous vectors, and then employ trained neural network models to assess compliance. These methods can optimize and improve performance with new data and regulations but may lose critical structured information and context, affecting compliance checking accuracy.

To integrate the advantages of the aforementioned methods, we design a compliance checking framework based on Retrieval-Augmented Generation (RAG), which combines structured knowledge with the parametric knowledge of Large

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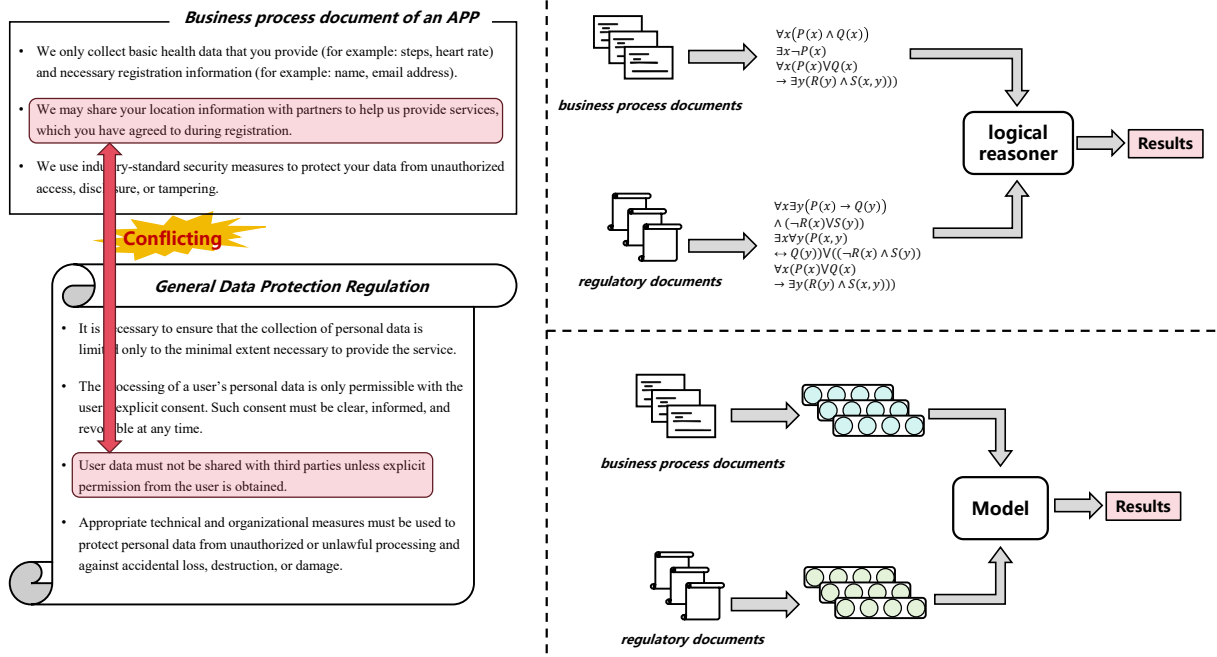


Figure 1: The left depicts a schematic diagram of the text-based compliance checking task, while the upper right corner illustrates compliance checking methods grounded in pure logical reasoning. Conversely, the lower right corner portrays compliance checking methods rooted solely in embedding techniques.

Language Models (LLMs) to achieve enhanced compliance checking. Our framework consists of three layers: the static layer, the dynamic layer, and the computational layer. In the **static layer**, we utilize conventional knowledge graphs to represent factual knowledge, including entities or concepts and their relationships, as well as term definitions. The **dynamic layer** comprises two sub-modules: a regulatory information module based on an eventic graph and a business process information module based on chunk vectors. In the **computational layer**, we retrieve relevant information from the static and dynamic layers to guide the LLM in generating compliance reasoning results. Additionally, to construct the eventic graph about regulatory information, we propose an unsupervised information extraction algorithm centered on deontic propositions.

We conducted extensive experiments on four existing compliance checking datasets. The experimental results demonstrate that the proposed framework significantly outperforms other baselines. Additionally, the experiments indicate that our framework exhibits greater flexibility compared to other supervised approaches, enabling rapid and unsupervised adaptation to other target domains. Furthermore, the experiments show that our framework is more adept at identifying compliance conflicts that require global reasoning, a task that even

the latest LLMs struggle to accomplish.

2 Related works

There are currently two primary text-based compliance checking methods: logic-based reasoning and semantic embedding-based methods (Hashmi et al., 2018; Ly et al., 2015).

A logic-based reasoning approach first formalizes the semantics of business processes and regulatory requirements, then performs explicit reasoning on this formalization to assess the compliance of business processes. For instance, some studies focus on converting natural language texts into first-order predicate logic expressions, followed by explicit logical reasoning using tools like Prolog or CLIPS (Governatori et al., 2006; Zhang and El-Gohary, 2017). Additionally, some research employs Petri nets—a widely used formal modeling language—to model business processes, enabling reasoning and verification of compliance by examining the transitions and states within the Petri net models (Rojas et al., 2016; Rozinat and Van der Aalst, 2008). There are also industry-specific formal modeling and reasoning methods, such as Building Information Modeling (BIM) for checking architectural compliance (Guo et al., 2021; Zheng et al., 2024) and Formal Contract Language (FCL) for verifying contract consistency.

The method based on semantic embeddings first employs low-dimensional continuous embedding vectors to represent the semantics of business processes and regulatory requirements, and then infers compliance implicitly through vector calculations (Saeidi et al., 2021). For instance, (Aires et al., 2018) embedded texts using sent2vec and then calculated their semantic compliance relationships using offset. (Aires and Meneguzzi, 2021) used convolutional networks to extract key semantic features from business process texts and regulatory texts, and determined the compliance of a business process through a classification head. (Huang et al., 2024) applied multi-task deep learning to simultaneously focus on both global and local semantic information in texts, thereby accurately capturing the compliance relationships between texts.

The advantage of logic-based methods is their ability to accurately capture structured relationships between business processes and regulations, with good interpretability. However, they lack scalability and rely heavily on domain experts, limiting adaptability to evolving regulatory requirement. In contrast, semantic embedding-based methods are flexible and less dependent on experts, but they lose structured process information and global context, reducing accuracy and interpretability. Our framework combines the strengths of both approaches. Unlike previous studies (Guo et al., 2021; Zheng et al., 2022), we integrate eventic graphs with LLMs using the RAG paradigm, a novel approach in compliance checking.

3 Framework

We design a static layer and a dynamic layer for store information. Additionally, we use a computational layer to retrieve knowledge and drive the reasoning process for compliance checking within the LLM. Therefore, our framework consists of three layers, as illustrated in Figure 2: the static layer for storing factual knowledge, the dynamic layer for storing regulatory information and business process information, and the computational layer responsible for global scheduling and executing compliance reasoning. The construction or computation process of each layer will be introduced in detail below.

3.1 Static layer

While entities, actions, states, and deontic propositions plays a central role in compliance check-

ing tasks, static factual knowledge is equally indispensable. For instance, a simple factual statement like “Huawei Technologies Co., Ltd. is located in Guangdong Province” provides crucial information that allows us to infer that the information technology regulations in Guangdong Province are binding on Huawei Technologies Co., Ltd.

We store three types of factual knowledge in the static layer: entity-centric, concept-centric, and term definition-centric knowledge. Given the widespread use of entity- and concept-centric knowledge graphs in the industry, we opt for existing open-source resources. We select CN-Dbpedia (an entity-centric knowledge graph) and OpenConcepts (a concept-centric knowledge graph) for the Chinese datasets. Besides, we chose Dbpedia (an entity-centric knowledge graph) and ConceptNet (a concept-centric knowledge graph) for the English datasets.

To obtain the definitions of terms, we trained a model for joint extraction of terms and their definitions. This model employs BigBird as the backbone, with a CRF layer appended to capture global label dependencies within the sequence. BigBird was chosen because most terms and their definitions are not confined to a single sentence but span across entire paragraphs. Therefore, we used paragraphs as input units and leveraged the BigBird model, which supports long text inputs. We annotated 5,500 term-definition pairs to serve as the training and development sets. The trained model, denoted as the function M_{exp} , is used to extract terms and their definitions from domain-specific regulatory documents $\mathcal{R} = \{r_1, r_2, \dots, r_{|\mathcal{R}|}\}$ and business process documents $\mathcal{B} = \{b, b_2, \dots, b_{|\mathcal{B}|}\}$, as shown in Equation 1.

$$\begin{aligned} M_{exp} : (\mathcal{R} = \{r_1, r_2, \dots, r_{|\mathcal{R}|}\}, \\ \mathcal{B} = \{b, b_2, \dots, b_{|\mathcal{B}|}\}) \mapsto ExpKnow \end{aligned} \quad (1)$$

The extracted terms and their corresponding definitions are stored in an knowledge base referred to as TDKnow. Ultimately, we integrate TDKnow with CN-DBpedia and OpenConcepts to form a comprehensive static factual knowledge graph for compliance checking, denoted as $\mathcal{G}_{static}$. The structural diagram of model M_{exp} , specific training details, and the annotation process of the term-definition pairs can be found in Appendix A.

3.2 Dynamic layer

The dynamic layer is utilized for storing business process information and regulatory information.

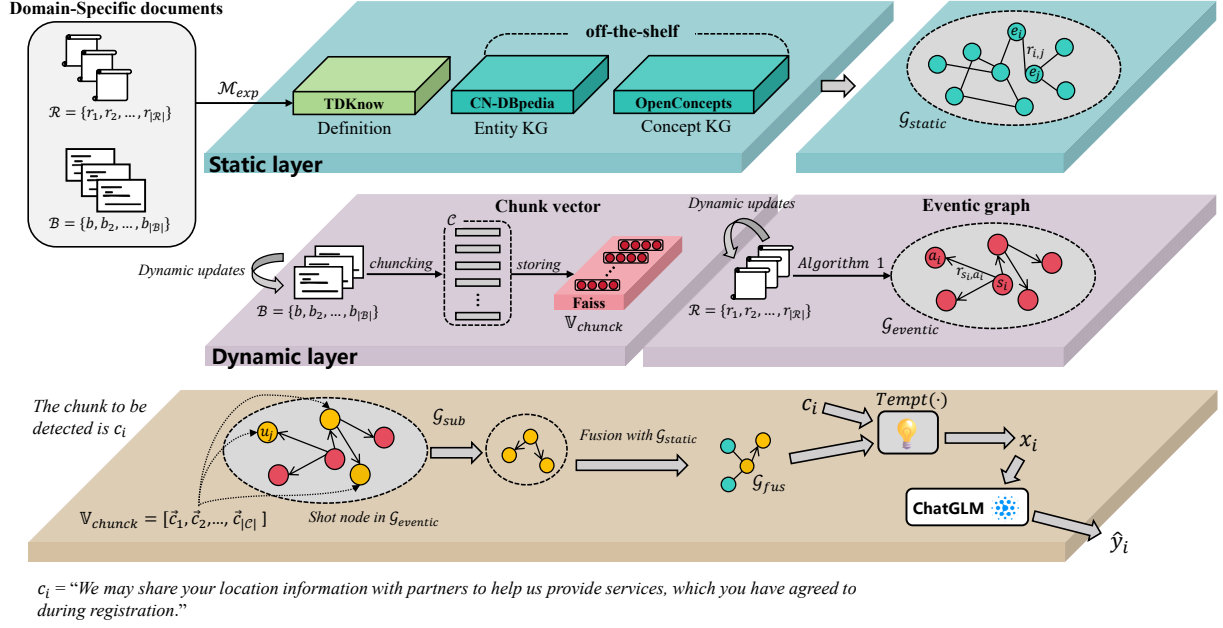


Figure 2: RAG-based compliance checking framework. This framework comprises three layers: the upper static layer (blue) stores static factual knowledge; the middle dynamic layer (purple) holds changing regulatory and business process information; the lower computing layer (yellow) retrieves pertinent knowledge to infer compliance outcomes.

We employ an eventic graph to describe regulatory information, as such information pertains not to entities and their relationships, but rather to knowledge regarding agents, actions, states, and deontic propositions. In addition, we use chunk vectors to represent business process information to facilitate vectorized computations. The following sections provide an introduction to these two components respectively.

Part 1: Eventic graph for regulatory information. The eventic knowledge graph focuses on describing the relationships between subjects, actions, states, and events, which aligns with the perspective used in regulatory information descriptions. Therefore, we construct an eventic graph capable of describing regulatory information, referred to as $\mathcal{G}_{eventic}$. Based on the characteristics of regulatory information and theories related to deontic propositions, we define the ontology for $\mathcal{G}_{eventic}$, which includes six entity types and seven relationship types, as detailed in Appendix B. Unlike conventional knowledge graphs where nodes are predominantly entity names, the nodes in $\mathcal{G}_{eventic}$ mainly describe actions and states, resulting in longer node texts. Consequently, standard named entity recognition and event extraction methods are ineffective in extracting the structured knowledge required for constructing $\mathcal{G}_{eventic}$. To address this issue, we propose an information extraction algorithm cen-

tered on deontic propositions, utilizing the API of a commercial LLM, as illustrated in Algorithm 1. To avoid the need for labor-intensive data annotation, Algorithm 1 is entirely unsupervised.

The core objective of Algorithm 1 is to extract regulatory information centered around deontic propositions from regulatory documents. The input to this algorithm is a collection of regulatory documents within a specific domain, denoted as $\mathcal{R} = \{r_1, r_2, \dots, r_{|\mathcal{R}|}\}$, and the output is a set of triples $\mathcal{E}$. In line 3, the algorithm aims to extract all agents subject to deontic constraints from the entire set of regulatory documents. Lines 4 through 8 are designed to extract deontic words from each paragraph based on the identified agents. According to our observations, nearly all actions or states under deontic constraints follow immediately after deontic words. Consequently, lines 8 through 11 of the algorithm extract these actions and states directly based on the predefined rules. The instruction templates for lines 3 and 8, $Tempt_1$ and $Tempt_2$, are provided in Appendix C. Finally, the triples centered on deontic propositions are stored in the set $\mathcal{E}$.

Part 2: Chunk vectors for business process information. To facilitate the computation of semantic vectors, we segment the business process into chunks and embed them as semantic vectors using the SBERT model proposed by (Zhao et al.,

Algorithm 1 Deontic Proposition-centered Information Extraction Algorithm based on LLM

INPUT: Regulatory document set $\mathcal{R} = \{r_1, r_2, \dots, r_{|\mathcal{R}|}\}$
OUTPUT: Set of triples $\mathcal{E}$

- 1: **Initialize** an empty set $\mathcal{E} = \{\}$
- 2: **for** each r_i in $\mathcal{R}$ **do**
- 3: $\mathcal{A}_{\text{agent}} \leftarrow \text{LLM}(\text{Tempt}_1(r_i))$ // Extract all agents from r_i via the LLM
- 4: Divide r_i to obtain sequence P_i of paragraphs.
- 5: **for** each p_j in P_i **do**
- 6: Match each *agent* in $\mathcal{A}_{\text{agent}}$
- 7: **if** *agent* in p_j **then**
- 8: $dword \leftarrow \text{LLM}(\text{Tempt}_2(\text{agent}, p_k))$ // Extract deontic word
- 9: $s \leftarrow$ End index of string *dword*
- 10: $e \leftarrow$ End index of string p_j
- 11: $action \leftarrow p_k[s : e]$
- 12: Add (*agent*, *dword*, *action*) into $\mathcal{E}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathcal{E}$

2019), storing these vectors in the Faiss (Qin et al., 2020) vector database. This process is illustrated in Equations 2-3.

$$\begin{aligned} \text{chunk} : \mathcal{B} &= \{b_1, b_2, \dots, b_{|\mathcal{B}|}\} \\ \mapsto \mathcal{C} &= \{c_1, c_2, \dots, c_{|\mathcal{C}|}\} \end{aligned} \quad (2)$$

$$\text{SBERT} : \mathcal{C} \mapsto \mathbb{C} = [\vec{c}_1, \vec{c}_2, \dots, \vec{c}_{|\mathcal{C}|}] \quad (3)$$

where $\mathcal{C}$ represents the set of chunks, and $\mathbb{C}$ is the matrix composed of the semantic vectors of these chunks.

3.3 Computing layer

The purpose of this layer is to retrieve knowledge from both the static and dynamic layers, thereby guiding the LLM to generate accurate and reliable compliance checking results. Initially, we allow the vectors in $\mathbb{C}$ to sequentially match the knowledge graph $\mathcal{G}_{\text{eventic}}$. Specifically, for each vector $\vec{c}_i$, we calculate its semantic similarity with each node u_j in $\mathcal{G}_{\text{eventic}}$. If the semantic similarity exceeds the threshold λ , the node u_j is considered a hit. This process is illustrated in Equation 4.

$$f(u_j) = \begin{cases} \text{hit}, & \text{if } \cos(\vec{c}_i, \vec{u}_j) \geq \lambda \\ \text{miss}, & \text{else} \end{cases} \quad (4)$$

where $\vec{u}_j$ represents the embedding vector of node u_j (using SBERT for semantic embedding). All hit nodes in $\mathcal{G}_{\text{eventic}}$ are denoted as $\mathcal{G}_{\text{sub}}$, with $\mathcal{G}_{\text{sub}} \subset \mathcal{G}_{\text{eventic}}$.

Next, nodes from $\mathcal{G}_{\text{sub}}$ are matched with those in $\mathcal{G}_{\text{static}}$ to obtain the intersection set $\mathcal{P}$, as shown in Equation 5.

$$\mathcal{P} = \{p \in V_{\mathcal{G}_{\text{static}}} | p = s \in V_{\mathcal{G}_{\text{sub}}}\} \quad (5)$$

where $V_{\mathcal{G}_{\text{static}}}$ represents the set of nodes in $\mathcal{G}_{\text{static}}$, and $V_{\mathcal{G}_{\text{sub}}}$ represents the set of nodes in $\mathcal{G}_{\text{sub}}$.

Subsequently, the set of neighbor nodes $\mathcal{N}$ of the set $\mathcal{P}$ is obtained, as shown in Equation 6.

$$\mathcal{N} = \cup_{p \in \mathcal{P}} N(p) \quad (6)$$

where p represents a node in $\mathcal{P}$, and $N(p) = \{n \in V_{\mathcal{G}_{\text{static}}} | (p, n) \in E_{\mathcal{P}_{\text{static}}}\}$. Besides, $E_{\mathcal{G}_{\text{static}}}$ denotes the edge set of $\mathcal{G}_{\text{static}}$. Ultimately, the largest connected graph with $\mathcal{N}$ as its nodes is obtained, denoted as $\mathcal{G}_{\text{fus}}$. $\mathcal{G}_{\text{fus}}$ is a heterogeneous knowledge graph related to $\vec{c}_i$, encompassing both static and dynamic knowledge.

We wrap the current chunk c_i and its associated knowledge graph $\mathcal{G}_{\text{fus}}$ using the instruction template $\text{Tempt}_3(\cdot)$ to obtain the input x_i . The instruction template is illustrated in Appendix D.

Finally, the input x_i is provided to ChatGLM-3-6b, guiding it to generate compliance checking results with explanations, as shown in Equation 7. For the datasets in English, we employ the LLaMa-2 model.

$$\text{ChatGLM} : \text{Tempt}_3(c_i, \mathcal{G}_{\text{fus}}) \mapsto \hat{y}_i \quad (7)$$

where $\hat{y}_i$ represents the compliance checking results with explanations generated by the LLM.

4 Experimental settings

4.1 Datasets

To evaluate the effectiveness of our framework, we conducted experiments on the following four datasets: **1) EU2UK** is a dataset proposed by (Chalkidis et al., 2021) aimed at checking whether UK legislation complies with EU directives. This dataset is designed for document-level compliance checking, where the unit of compliance conflict is the entire document rather than individual sentences. **2) GDPR-13** is a dataset introduced by Liu et al. (Liu et al., 2021) used to analyze the compliance of privacy policies of applications with Chapter 13 of the GDPR. **3) CONTRACT** is a

dataset developed by (Aires et al., 2019) intended to assess contract consistency. **4) CSSCD** is a Chinese dataset proposed by (Huang et al., 2024) for analyzing compliance in social security operations. This is the only Chinese dataset among the four; the other three datasets are in English.

4.2 Baselines

We compare our framework with the following baselines: **1) Doc2Doc**, an information retrieval-based method focused on document-level compliance checking, which achieved the best performance on the EU2UK dataset (Chalkidis et al., 2021). **2) CLS** represents the semantic classification-based compliance checking method proposed by (Liu et al., 2021), where business sentences are categorized into predefined violation classes using machine learning or neural network classifiers. **3) Offsets**, originally proposed by (Aires et al., 2018) as a contract consistency method based on semantic embeddings, is also applicable to compliance checking tasks. **4) NeuralConflict** is a multi-task learning-based compliance checking method designed to analyze both local and global semantic compliance simultaneously (Huang et al., 2024).

Additionally, considering the similarity in objectives between compliance checking and textual entailment tasks, we also included the following baselines: **5) TER-PLM**, a method for textual entailment recognition that directly concatenates two sentences (with [SEP] as a separator) as input to language models (Wehnert et al., 2022). **6) TER-Inner**, an interaction attention-based textual entailment recognition method, which has demonstrated effectiveness across multiple datasets (Sun et al., 2017). **7) TER-GraphAtt**, a textual entailment recognition method based on graph attention networks, which has achieved state-of-the-art results on several datasets (Chen et al., 2019).

5 Results and analysis

5.1 Performance comparison with baselines

We first analyze the performance of the aforementioned baselines and our framework across four datasets. The compliance checking task is characterized by a severe imbalance in the number of positive and negative samples, with the majority of business processes being compliant, while non-compliant cases are relatively few. In light of this, we use the Matthews Correlation Coefficient (MCC) as the evaluation metric. MCC is a measure

for assessing the performance of binary classifiers, particularly effective in handling datasets with imbalanced sample distributions.

The experimental results are presented in Table 1. Upon examining the table, it is evident that our proposed framework demonstrates outstanding performance in compliance checking, which strongly attests to the framework’s effectiveness and advancement. Notably, on the document-level compliance checking dataset EU2UK, our framework exhibits a particularly significant performance advantage, surpassing the second-place method by 0.048. This indicates that our framework possesses a remarkable capability for conducting global compliance analysis and checking.

In the tests conducted on the non-document-level datasets GDPR-13, CONTRACT, and CSSCD, the performance of the text entailment recognition baselines surpassed that of the compliance checking baselines. This suggests that compliance checking methods based on semantic embeddings may potentially be replaced by other deep learning models with similar task objectives. However, our approach consistently outperformed all baselines across the four datasets. This is attributed to our framework’s reasoning process not only focuses on the semantic consistency between current business process sentences and regulatory statements but also emphasizes the comprehensive analysis of global regulatory information. Furthermore, the static and dynamic knowledge layers demonstrated unique and irreplaceable advantages, as further verified and analyzed in the experiments discussed in Section 5.2 and 5.3.

Table 1: Performance of our proposed framework and baselines on the four datasets. The evaluation metric used is the Matthews Correlation Coefficient (MCC).

Method	English			Chinese
	EU2UK	GDPR-13	CONTRACT	CSSCD
Doc2Doc	0.724	0.601	0.674	0.628
CLS	/	0.602	0.672	0.633
Offsets	/	0.584	0.648	0.611
NeuralConflict	/	0.610	0.661	0.643
TER-PLM	/	0.617	0.694	0.645
TER-Inner	/	<u>0.620</u>	0.712	<u>0.650</u>
TER-GraphAtt	<u>0.730</u>	0.619	<u>0.713</u>	0.648
Our framework	0.778	0.652	0.730	0.680

5.2 Ablation study

This section analyzes the contributions of each module within our framework. We sequentially remove the three knowledge graphs from the static layer until the entire static layer is eliminated. As

shown in Figure 3, the performance of our framework on all four datasets consistently declines with the successive removal of the knowledge graphs. This indicates that each knowledge graph within the static layer—namely, the entity graph, the concept graph, and the term definition graph—is indispensable for compliance checking within the framework.

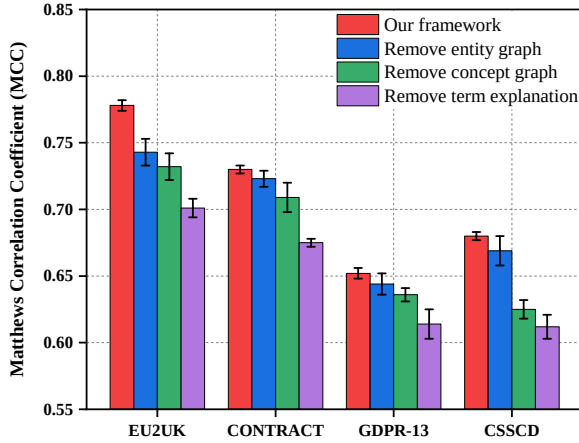


Figure 3: Results of the ablation study.

Furthermore, we observe that when all the knowledge graphs, i.e., the entire static layer, are removed, the compliance checking performance of our framework falls below that of other baselines. This suggests that factual knowledge is essential in compliance checking tasks.

5.3 Domain adaptability

We can apply the designed framework in a domain-independent manner, with only the term definition extraction module requiring domain-specific labeled data. To overcome this limitation, we leverage the In-Contextual Learning (ICL) capabilities of LLMs to reduce the dependence on domain-specific data. Specifically, after demonstrating the annotation patterns of terms and definitions from the source domain to the LLM, the model is then directed to perform unsupervised annotation on the target domain. The instruction template used to guide the LLM’s annotation is shown in Appendix E.

We use the CSSCD dataset as the source domain and the other three datasets as the target domains. The experimental results in Table 2 demonstrate that our framework maintains effective compliance checking performance even in cross-domain scenarios.

Table 2: The effectiveness of the term and explanation extraction in the domain transfer scenario.

Source domain	CSSCD	MCC
Target domains	EU2UK	0.743
	GDPR-13	0.603
	CONTRACT	0.714

5.4 Global information perception

To evaluate the framework’s capability in checking global compliance, we analyzed a specific case, as illustrated in Figure 4. In the figure, the business process sentence is “We will share your personal data with third parties as necessary,” while a regulatory requirement states, “The processing of personal data requires lawful basis.” Analyzing these two sentences alone reveals no compliance conflict. However, considering an additional regulatory requirement, “A user’s explicit consent is the only lawful basis for sharing data,” it becomes evident that there is a conflict between the business process and the regulatory requirements when both regulations are considered together.

We tested whether the baseline methods could identify this compliance conflict, several open-source LLMs were also included in the experiments. All traditional baseline methods failed to identify this compliance conflict. Among the LLM baselines, only MoonShot-v1-128k and GPT-4 correctly identified this conflict. We hypothesize that this is because the pre-training data for these two LLMs included the General Data Protection Regulation and the business process of this APP.

5.5 Analysis of influencing factors

This section analyzes the factors that may influence the performance of our framework. First, we examine the impact of the sole hyperparameter, λ , which requires manual configuration. As illustrated in Figure 5(a), our framework performs optimally when λ is set to 0.7 or 0.8. When λ is set to 0.9, the compliance checking capability of the framework declines sharply. We attribute this to the excessively high threshold prevents c_i from connecting to a sufficient number of nodes, thereby limiting the structured knowledge available for LLM reasoning.

Subsequently, we replaced the language model SBERT in Section 3.2 with BERT, RoBERTa, XLNet, ELECTRA, and DistilBERT, respectively. As shown in Figure 5(b), substituting different language models had minimal impact on the frame-

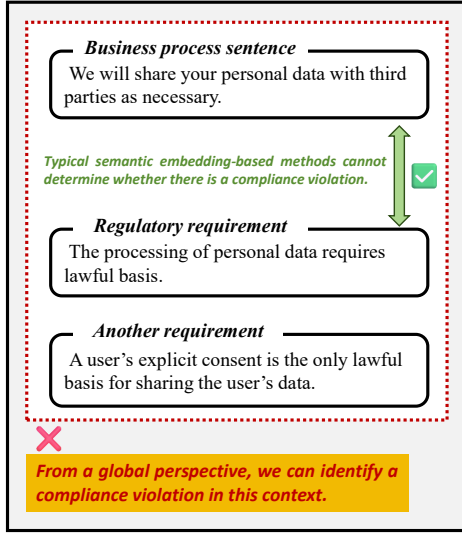


Figure 4: The business process statement in the diagram does not conflict with the first regulatory requirement. However, when both regulatory requirements are considered simultaneously, the business process is in violation of the regulations.

work's performance.

Finally, we replaced ChatGLM-3-6b with other LLMs. As depicted in Figure 5(c), ChatGLM demonstrates the best performance on the Chinese dataset, while LLaMa-2 achieved the best results on the English datasets.

6 Conclusion

We proposed a compliance checking framework based on Retrieval-Augmented Generation (RAG), which leverages the strengths of both embedding-based and logic-based compliance checking methods. The framework consists of three layers: a static layer that stores factual knowledge, a dynamic layer that holds regulatory information and business process data, and a computational layer responsible for knowledge retrieval and guiding the reasoning of Large Language Models (LLMs). Experimental results demonstrate that our framework achieves state-of-the-art performance across four compliance checking datasets. Moreover, our framework exhibits superior do-main adaptability and is capable of detecting potential compliance conflicts from a global perspective.

7 Limitations and potential solutions

Our framework relies on the retrieval of knowledge from both the static and dynamic layers. Despite carefully designing retrieval strategies and setting retrieval thresholds, the thresholds originally set

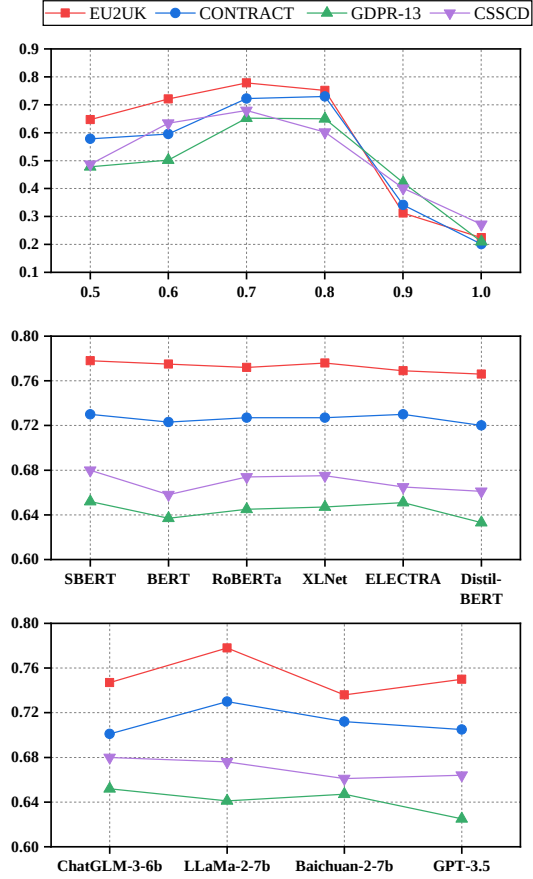


Figure 5: Subfigure (a) illustrates the impact of the hyperparameter λ on the framework, subfigure (b) demonstrates the influence of different encoder models on the framework, and subfigure (c) presents the effects of various LLMs on the framework.

may become invalid once the knowledge in the static and dynamic layers is updated, potentially interfering with the reasoning of the LLM. A potential future solution could involve designing an LLM agent-based knowledge retrieval mechanism, where agents with autonomous decision-making capabilities dynamically and intelligently retrieve the knowledge needed for LLM reasoning.

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A BigBird-based jointly extraction model for terms and explanations

Figure 6 illustrates the structure of the joint extraction model for terms and explanations. This is a typical model structure used for sequence labeling, where a CRF layer is added after BigBird to model dependencies between labels, thereby improving the global prediction accuracy of the label sequence. As shown in the figure, terms and their explanations are extracted simultaneously, ensuring the sharing of task information. BigBird allows for input of up to 8,000 tokens, making it capable of accepting input text in whole paragraph units.

We annotated 5,500 samples for the training and development validation of the model, with 4,000 used for training and 1,500 for development validation. First, we divided business process and regulatory documents from specific domains into paragraphs. Then, three junior annotators each annotated 2,000 samples, resulting in a total of 6,000 annotated samples. These 6,000 samples were then reviewed by two senior annotators, who removed 500 poorly annotated samples, leaving us with 5,500 high-quality annotated samples. We used Fleiss’ Kappa to evaluate the consistency between the two senior annotators during the review process. The formula for calculating Fleiss’ Kappa is as follows:

$$Kappa = 1 - \frac{P_o - P_e}{1 - P_e} \quad (8)$$

In this formula, P_o represents the observed proportion of agreement between the reviewers, while P_e represents the expected proportion of agreement based on random chance. We collected the annotation results of the two senior annotators for the 500 samples that were removed and calculated their consistency. By comparing their annotations, we obtained a Kappa value of 0.78, which indicates a high level of agreement between the two senior annotators during the review process.

During the model training process, we set the maximum input length to 5,000, the batch size to 4, and the learning rate to $2e-5$. We used AdamW as the optimizer, with the maximum number of optimization epochs set to 20. Additionally, we implemented early stopping to prevent overfitting.

B Ontology of deontic proposition-centered eventic graph

We have defined the entity types and relationship types included in the ontology for the event graph $\mathcal{G}_{eventic}$. Table 3 presents the entity types along with corresponding examples, while Table 4 presents the relationship types and their examples.

C Templates for deontic proposition-centered information extraction

In line 3 of Algorithm 1, template $Tempt_1$ is used to guide the large model in extracting all agents from the given regulatory document. The template is as shown in Figure 7.

Table 3: Predefined entity types for eventic graph.

Entity type	Example
Organization	Huawei Technologies Co., Ltd
Person	Ren Zhengfei
Regulatory document	“Regulation on the Internet Protection of Minors”
Category	Data Security, Protection of minors
Action	Share user’s data with third parties
State	Stored on an isolated server

Table 4: Predefined relation types for eventic graph.

Relation type	Example
Publish	“Internet Information Office of China”, Publish, “Regulations on Ecological Governance of Network Information Content”
WorkFor	“Ren Zhengfei” WorkFor “Huawei Technologies Co., Ltd”
Duty	“Information technology companies” Duty “Ensure users’ personal data is limited only to the minimal extent necessary to provide the service”
Prohibited	“APP service provider” Prohibited “Share user’s data with third parties”
Right	“Shenzhen Market Supervision and Administration Bureau” HaveRight “Penalties for violating companies”
ClassifiedTo	“Regulation on the Internet Protection of Minors” ClassifiedTo “Information security”
Cite	“Regulation on the Internet Protection of Minors” Cite “Minors Protection Act”

In line 8 of Algorithm 1, template $Temp_2$ is used to guide the large model in predicting the moral words based on the current paragraph and the subject it contains. The template is as shown in Figure 8.

D Template for wrapping chunks and their related knowledge

Template for wrapping chunks and their related knowledge is as shown in Figure 9.

E The instruction template used to guide the LLM’s annotation

The instruction template used to guide the LLM’s annotation is shown in Figure 10.

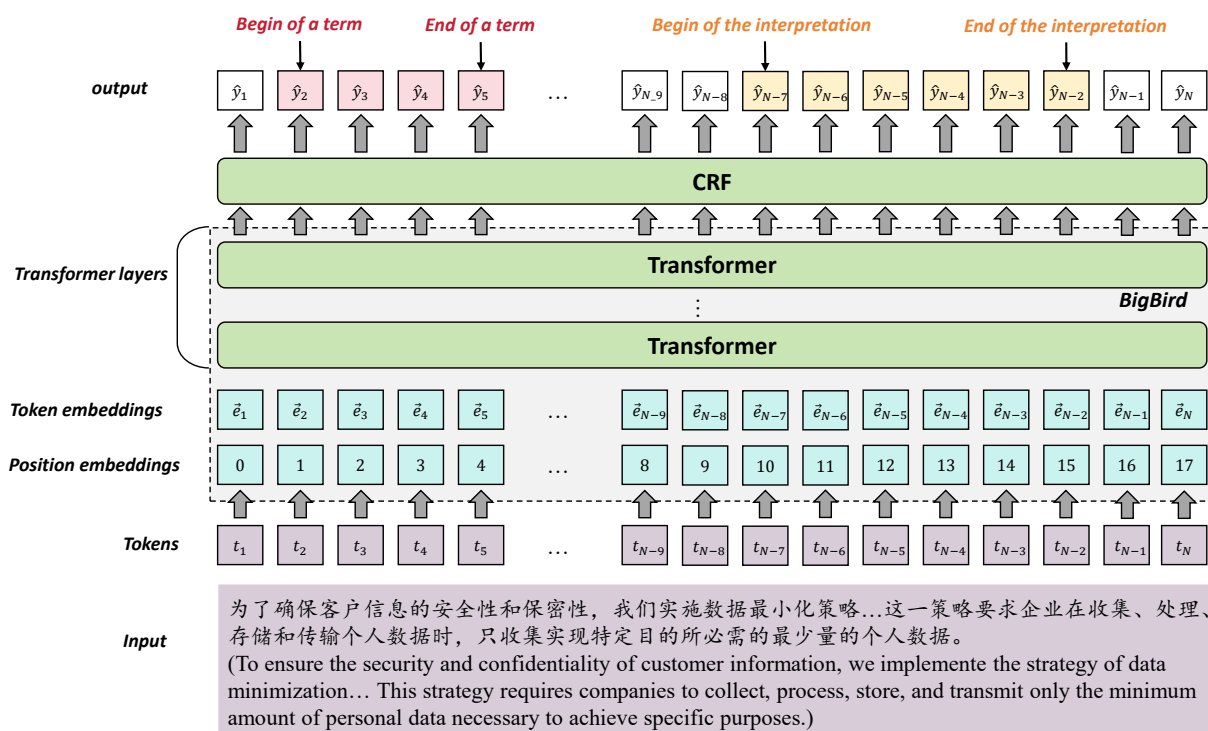


Figure 6: Structure of the jointly extraction model for terms and explanations.

请仔细阅读附件中的监管文件，并从中识别出所有受道义约束的主体。这些主体可能包括但不限于组织、公司、政府部门、非营利组织以及个人等。注意寻找文档中使用的词汇，如“必须遵守”、“有义务”、“应当”等，这些通常指示了某种形式的约束或责任。请将识别出的主体及其相关的道义约束以清晰的列表形式返回。准确性和完整性非常重要，请确保不遗漏任何相关的信息。

例如，如果文档中有句子“公司必须保护客户的个人信息”，那么“公司”就是一个受约束的主体，其道义约束是“保护客户的个人信息”。

[Regulatory document attachment]

Figure 7: Template $Tempt_1$ is used to guide the large model in extracting all agents from the given regulatory document.

任务：请仔细阅读以下段落，识别出与受道义约束的主体相关的道义词，并为每个预测的道义词提供简短的解释。道义词是指表达某种道德或法律义务、责任或规范的词汇，如“必须”、“应当”、“禁止”、“责任”等。

段落：
[Insert paragraph text]

受道义约束的主体：
[Insert agent name]

示例：段落文本：“医疗机构必须保护患者的隐私。”主体：“医疗机构”预测道义词：“必须”解释：在这个段落中，“必须”表达了医疗机构有法律和道德上的义务来保护患者的隐私。

[Provide your prediction and explanation here]

Figure 8: Template $Tempt_2$ is used to guide the large model in predicting the moral words based on the current paragraph and the subject it contains.

给定文本(Given text): $[c_i]$

请判断以下知识片段中是否存在与 $[c_i]$ 的语义相互矛盾的情况:

(Please evaluate whether there is a semantic contradiction between the following knowledge fragments and c_i .)

[Head_Entity_1] [Relation_1] [Tail_Entity_1]

[Head_Entity_2] [Relation_2] [Tail_Entity_2]

...

[Head_Entity_n] [Relation_n] [Tail_Entity_n]

请依次分析每个知识片段与 c_i 之间是否矛盾, 并给出理由。

(Analyze each knowledge fragment in sequence to determine whether it contradicts c_i , and provide the reasoning.)

如果不存在任何知识片段与 c_i 矛盾, 则返回给我<合规性检测通过>

(If no knowledge fragment contradicts c_i , return <Compliance Check Passed>.)

如果存在知识片段与 c_i 矛盾的情况, 则返回给我<合规性检测不通过>, 并在{}中返回产生矛盾的知识片段以及理由。

(If any knowledge fragment contradicts c_i , return <Compliance Check Failed>, and include the conflicting knowledge fragment and the reasoning within {}.)

Figure 9: Template for wrapping chunks and their related knowledge.

你将参与一个术语及其解释的标注任务, 该任务旨在帮助我们理解和抽取特定领域的知识。

(You will participate in an annotation task involving terms and their definitions. This task aims to assist us in understanding and extracting knowledge specific to a particular domain.)

在开始之前, 我将向你展示一些来自源域的示例, 这些示例包含了术语及其解释。请仔细阅读并理解这些标注模式。

(Before you begin, I will show you some examples from the source domain, which include terms and their definitions. Please read and understand these annotation patterns carefully.)

[Example]

{文本(Text): [Paragraph_1],

术语(Term): 社会保险,

定义(Definition): 社会保险是一种由政府主导的社会保障制度, 旨在为个人提供养老、医疗、失业、工伤和生育等方面的保障。}

{文本(Text): [Paragraph_2],

术语(Term): 个人养老金账户,

解释(Definition): 个人养老金账户中的资金来源于个人和单位的定期缴费, 这些资金将用于参保人员退休后的养老金发放。}

[Target Domain Task]

现在, 我需要你将这种理解应用到目标域上。请在没有直接标注数据的情况下, 自动识别和标注目标域中的术语及其解释。

(Now, I need you to apply this understanding to the target domain. Please automatically identify and annotate the terms and their explanations in the target domain without directly annotated data.)

[Task Instructions]

1. 仔细阅读目标域中的文本。(Carefully read the text in the target domain.)

2. 识别出文本中的专业术语。(Identify the specialized terms in the text.)

3. 每个识别出的术语都对应一个清晰、准确的解释。(Each identified term should correspond to a clear and accurate definition.)

Figure 10: The instruction template used to guide the LLM's annotation.